Smart Home Energy: Sample Portfolio of 50 Units

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Prepared for a property developer with a 50-unit development (simulated client). Prepared by the Data Science Internship.

Summary

A random sample of 50 units uses almost exactly the same energy as the 500-home benchmark, and two appliances drive nearly half of its estimated cost.

- **Typical portfolio:** one measurement uses 1.50 kWh on average in both the portfolio and the benchmark, an estimated $0.495 (about 50 cents) at a flat rate of 33 cents per kWh.
- **Where the cost is:** Air Conditioning (23.8%) and Heater (22.7%) make up 46.5% of estimated cost. Every other appliance is 7% to 8%, and the Fridge is 1.9%.
- **No unit stands out:** for every appliance the portfolio is within about 2% of the benchmark, and the highest unit is 8.6% above the average.
- **Recommendation:** focus efficiency effort on Air Conditioning and Heater, use the dashboard to compare any unit with the benchmark, and test the tool on real meter data.

Note: the data appear to be simulated, so the figures show how the method works, not facts about real buildings.

The sections below describe the data

Read this part first. It explains what we worked with and where the data has limits. Technical terms are explained in the glossary in Appendix B.

The data and the sample

The benchmark is 500 smart homes; the portfolio is 50 of them, chosen at random.

- **Benchmark:** 500 homes, 10 types of appliances and about 100,000 measurements, from 1 January 2023 to 1 January 2024.
- **A measurement** is one recorded energy value (in kWh) for one appliance in one home at one moment. Each home has 157 to 245 measurements in the year, with a median gap of about 30 hours between two measurements of the same home. They are separate snapshots, not a continuous record.
- **Portfolio:** 50 homes chosen at random, and the list of homes is saved so the same sample can be used again. They have 10,158 measurements in total.

Limits of the data

The data have limits that you should keep in mind when you read the results.

- **Simulated data:** the data appear to be simulated, not recorded in real homes. Energy use does not change with season, time of day or outdoor temperature, and the homes are very alike. Appliance type is the only clear difference, so this report shows how the method works. In real homes, season, time of day and weather would almost certainly matter.
- **Snapshots:** each home has only about 200 measurements in the year, so costs are shown per measurement, not as a monthly or yearly bill.
- **No home details:** the data hold no floor area or number of occupants, so large and small homes cannot be compared fairly and cost per person cannot be shown.
- **Example cost:** the electricity rate is an example for one region and plan, and only the usage charge is included.
- **No natural groups:** we checked whether the 500 homes split into separate kinds, for example one group of homes that use a lot of electricity and another group of homes that use very little. They do not: the homes all look alike, with no clear line between one kind and another. So, every unit is compared with all 500 homes, and the same advice applies to every unit. With real data we can run the check again, because real homes may split into kinds.

What it means for your portfolio

The sections below show where your cost is, how your units compare with the 500 homes, and what we recommend.

Energy use and cost by appliance

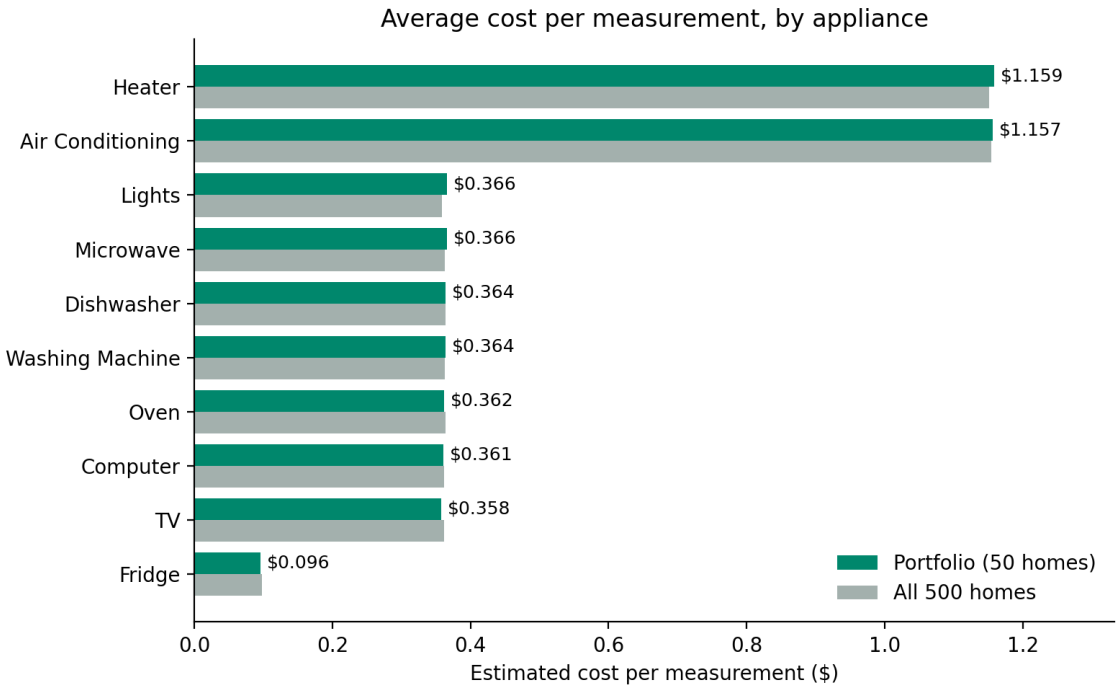
Air Conditioning and Heater use about 3.5 kWh per measurement, roughly three times a typical appliance and twelve times the Fridge.

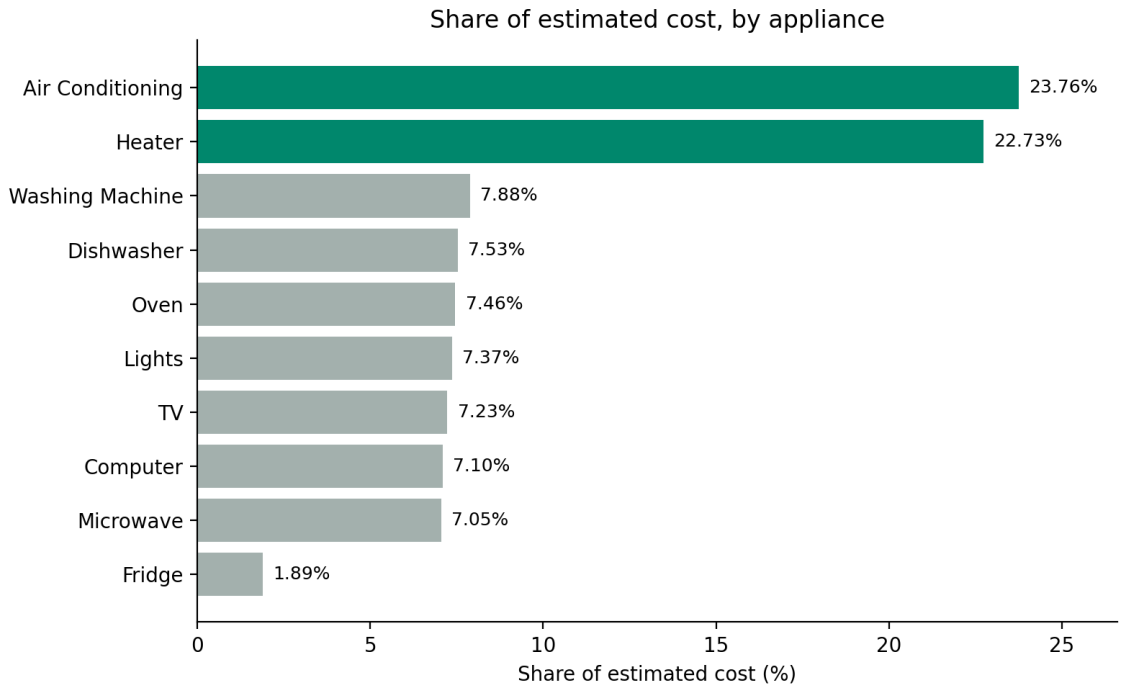
Appliance	Portfolio avg kWh	All homes avg kWh	Difference from all homes	Portfolio cost ($)	All homes cost ($)	Share of portfolio cost
Air Conditioning	3.505	3.500	+0.1%	1.157	1.155	23.76%
Heater	3.514	3.487	+0.8%	1.159	1.151	22.73%
Washing Machine	1.103	1.101	+0.1%	0.364	0.363	7.88%
Dishwasher	1.103	1.103	-0.1%	0.364	0.364	7.53%
Oven	1.098	1.103	-0.4%	0.362	0.364	7.46%
Lights	1.108	1.087	+1.9%	0.366	0.359	7.37%
TV	1.086	1.097	-1.0%	0.358	0.362	7.23%
Computer	1.094	1.095	-0.2%	0.361	0.362	7.10%

Appliance	Portfolio avg kWh	All homes avg kWh	Difference from all homes	Portfolio cost ($)	All homes cost ($)	Share of portfolio cost
Microwave	1.110	1.101	+0.8%	0.366	0.363	7.05%
Fridge	0.292	0.298	-2.1%	0.096	0.098	1.89%

How to read the table:

- Average kWh: the average energy used in one measurement.
- Measurement: one recorded energy value (kWh) for one appliance in one home. The data does not say how long each one covers.
- Cost: energy times 33 cents per kWh (rounded from the AER's 33.14 cents).
- Difference from all homes: how far the portfolio is above or below the average of all 500 homes, in per cent. For example, +1.9% for Lights means the portfolio uses 1.9% more energy for lights.
- Share of portfolio cost: how much of the portfolio's total cost comes from each appliance, in per cent. For example, 23.76% for Air Conditioning means that about 24 of every 100 dollars of estimated cost come from the air conditioner.

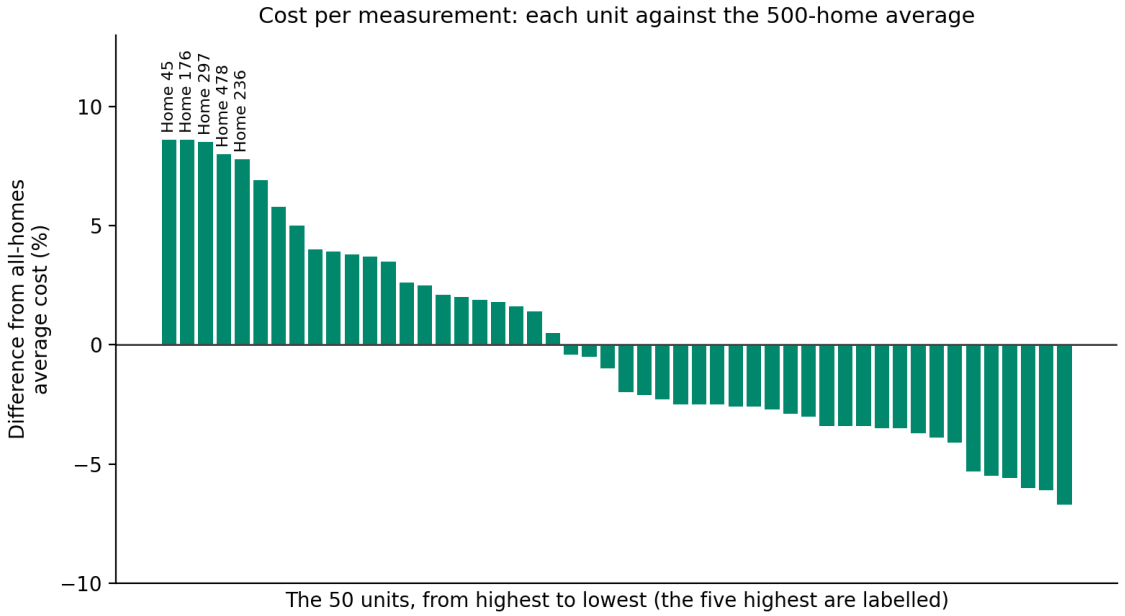




How the portfolio compares with the 500 homes

For every appliance the portfolio is within about 2% of the 500-home average, from 2.1% below (Fridge) to 1.9% above (Lights), so it looks typical.

Figure below shows all 50 units. Every unit is between 6.7% below and 8.6% above the 500-home average, with 22 units above it and 28 below, so no unit is far from the middle.



The five units with the highest average cost per measurement are shown below. They are 7.8% to 8.6% above the all-homes average of $0.495.

Home	Measurements	Avg kWh	Avg cost ($)	Difference from all-homes average
176	207	1.630	0.538	+8.6%
45	217	1.629	0.538	+8.6%
297	183	1.628	0.537	+8.5%
478	191	1.620	0.535	+8.0%
236	221	1.617	0.534	+7.8%

Three points to keep in mind when you read this table:

- **Not high consumers:** a gap of 7.8% to 8.6% is small. In any group of 50 homes this alike, some homes have to be at the top.
- **Position only:** the comparison shows where a unit stands, not whether an appliance is efficient or wasteful.
- **Averages, not totals:** some homes were measured more often than others, so comparing totals would not be fair.

Estimated cost

Cost is energy multiplied by a flat rate of 33 cents per kWh (rounded from the AER's 33.14 cents), and it is shown per measurement, not as a bill.

- **Rate:** the Australian Energy Regulator's Default Market Offer 2026-27, Ausgrid area (NSW), single/flat rate, all-day usage charge, including GST. The AER states that all its tariff caps include GST. The price was read on 30 October 2026. It is an example, and a client can enter their own rate in the dashboard.
- **Not included:** the daily supply charge ($1.66 per day), because the data hold energy per appliance only.
- **Per measurement:** each home has only about 200 measurements, a median of 30 hours apart, so a monthly or yearly bill cannot be calculated reliably.
- **Result:** the portfolio's average estimated cost is $0.495 per measurement, the same as the benchmark's $0.495.
- **Illustration, not a forecast:** if Air Conditioning and Heater each used 10% less energy, the portfolio's estimated cost per measurement would fall by about 4.6%, around 2 cents.

Using the dashboard

The dashboard repeats this comparison for any unit or group of units, and the views are provided as a PDF with this report.

- **Overview:** average energy by appliance for the homes you pick. Choose All for the whole benchmark or select your own units.

- **Household profile:** one unit against the average home, appliance by appliance. The left chart shows how far the unit is above or below average. The right chart shows how many of every 100 homes use less energy than this unit, and the dashed line at 75 marks the top quarter of homes.
- **Appliance summary:** the same comparison as a table, with a Low, Medium or High label for each appliance, ready to hand to an owner or tenant.
- **Cost:** energy converted into dollars at an electricity rate you can change, with cost by appliance and the share from Air Conditioning and Heater.
- **How to read this dashboard:** plain-language definitions of every term and data.

A high position is a reason to look, not proof of waste. In this data the homes are very alike, so a small gap can look large in a ranking.

Recommendations

We recommend three actions, in this order. The first two tell you where to focus. The third makes the numbers reliable enough to act on.

1. **Make Air Conditioning and Heater the first priority in your efficiency programme.**
    - **Why:** together they are 46.5% of the portfolio's estimated cost. Each uses about 3.5 kWh per measurement, roughly three times a typical appliance. Every other appliance is 7% to 8%, and the Fridge is 1.9%.
    - **First steps:** get every air conditioner and heater serviced, check their temperature settings and timers, and give tenants simple tips. For example: set heating to 18 to 20 °C in winter and cooling to 25 to 27 °C in summer, because each extra degree adds about 5% to 10% to energy use (Australian Government, YourHome); use a timer so the unit is off when no one is home; and clean the filters regularly.
    - **What you get:** your effort goes to the appliances that cost the most. For example, if these two appliances used 10% less energy, the estimated cost would fall by about 4.6%, roughly 2 cents for every 50 cents.
2. **Use the dashboard as a regular check. A high position is a reason to look, not proof of waste.**
    - **Why:** the dashboard compares each unit with the 500 homes, one appliance at a time. In this sample no unit stands out: all 50 are between 6.7% below and 8.6% above the average.
    - **When to visit a unit:** each quarter, open the Household Profile page for each unit. Visit the unit if one appliance shows both of these: (1) its bar on the right-hand chart is above the dashed line at 75, which means it uses more than 75 of every 100 homes, and (2) its bar on the left-hand chart is above the dashed line at +20%, which means it uses more than 20% more than the average home. If only one is true, check again next quarter.
    - **Where to start:** on the Appliance Summary page, start with units that have two or more appliances marked High. The Cost page shows which units cost the most in dollars.
    - **Why these numbers:** the dashed line at 75 is already on the dashboard, and 20% is well above the usual gap between homes in this data (5% to 11%). Treat both as starting points and adjust them on real data.
    - **What you get:** one consistent way to decide which units to inspect first.
3. **Connect your own data before you act on individual units.**

- **Why:** this data is simulated, and each home has only about 200 measurements in a year. That is too little to see real differences between units, seasons or times of day, or to work out a yearly cost.
- **What we need from you:** (1) meter readings for each unit, with dates and times at regular intervals, by appliance if possible; (2) the floor area, number of bedrooms and number of occupants of each unit; (3) your electricity plan and region.
- **What you get:** the cost per month and per year for each unit; how energy use changes with the seasons and the time of day; the cost per person and per square metre, so large and small units compare fairly; and a ranking you can act on.
- **How to start:** begin with a pilot on 10 to 20 units that have smart meters, then extend to the whole portfolio.

Next steps

The findings illustrate the method on simulated data and should be confirmed on real meter data.

Next steps for your portfolio:

1. Share your meter data by home and appliance, with dates, floor area and number of occupants.
2. Tell us your electricity plan and region, so costs use your own rate.
3. Repeat this review on your real data, adding monthly and yearly cost and cost per person or per square metre.

Appendix

The appendix holds the unit table, a glossary and the method and sources.

A. Unit-by-unit table. One row per portfolio unit (50 rows), taken from the file portfolio50_by_home.csv: home, number of measurements, average kWh, average cost and difference from the all-homes average. Units are sorted by home number.

Home	Measurements	Avg kWh	Avg cost ($)	Difference from average
32	228	1.493	0.493	-0.5%
34	199	1.525	0.503	+1.6%
40	200	1.586	0.523	+5.8%
41	201	1.530	0.505	+2.0%
44	192	1.418	0.468	-5.5%
45	217	1.629	0.538	+8.6%
60	206	1.462	0.482	-2.5%
82	208	1.553	0.513	+3.5%
87	200	1.463	0.483	-2.5%

Home	Measurements	Avg kWh	Avg cost ($)	Difference from average
93	192	1.449	0.478	-3.4%
110	215	1.538	0.507	+2.5%
136	201	1.574	0.520	+5.0%
176	207	1.630	0.538	+8.6%
192	202	1.469	0.485	-2.0%
198	172	1.456	0.480	-2.9%
200	195	1.448	0.478	-3.5%
212	197	1.449	0.478	-3.4%
213	207	1.539	0.508	+2.6%
217	216	1.603	0.529	+6.9%
222	213	1.466	0.484	-2.3%
236	221	1.617	0.534	+7.8%
240	203	1.448	0.478	-3.5%
243	197	1.410	0.465	-6.0%
262	222	1.455	0.480	-3.0%
269	217	1.448	0.478	-3.4%
297	183	1.628	0.537	+8.5%
307	198	1.461	0.482	-2.6%
310	213	1.461	0.482	-2.6%
320	204	1.409	0.465	-6.1%
334	202	1.529	0.505	+1.9%
339	203	1.527	0.504	+1.8%
341	222	1.520	0.502	+1.4%
346	188	1.494	0.493	-0.4%
350	193	1.421	0.469	-5.3%
354	189	1.438	0.475	-4.1%
367	181	1.556	0.514	+3.8%

Home	Measurements	Avg kWh	Avg cost ($)	Difference from average
372	222	1.442	0.476	-3.9%
373	226	1.558	0.514	+3.9%
390	213	1.400	0.462	-6.7%
392	197	1.460	0.482	-2.7%
394	225	1.416	0.467	-5.6%
404	177	1.560	0.515	+4.0%
418	198	1.444	0.476	-3.7%
431	197	1.462	0.482	-2.5%
440	201	1.556	0.513	+3.7%
445	206	1.485	0.490	-1.0%
451	191	1.508	0.498	+0.5%
478	191	1.620	0.535	+8.0%
482	205	1.531	0.505	+2.1%
494	205	1.469	0.485	-2.1%

B. Glossary

Term	Meaning
Appliance	A device that uses electricity, such as the air conditioner, heater, fridge or TV
Unit	One home or apartment in the portfolio
Portfolio	The group of 50 units reviewed in this report
Benchmark	The average of all 500 homes, used as the point of comparison
Measurement	One recorded energy value (kWh) for one appliance in one home. The data does not say how long each one covers.
Snapshot	A single measurement taken at one moment, not a continuous recording. Each home has about 200 in the year
kWh (kilowatt-hour)	The unit of energy shown on an electricity bill

Term	Meaning
Average	The total divided by the number of values. Average kWh per measurement is the average of all measurements
Median gap	The middle value of the gaps between measurements: half the gaps are shorter and half are longer. Here it is about 30 hours
Average cost per measurement	The cost of the energy in one measurement, on average (average kWh per measurement multiplied by the electricity rate). Use it to compare homes and appliances, not as a bill
Difference from all homes	How far a unit or the portfolio is above or below the 500-home average, in per cent. +10% means 10% more
Share of portfolio cost (cost share)	The part of the portfolio's estimated cost that comes from one appliance, in per cent
Position (percentile)	Where a unit stands among the 500 homes. Uses more than 75% means 75 of every 100 homes use less energy than the unit. Statisticians call this a percentile
Top quarter	The 25 of every 100 homes that use the most energy for an appliance, which means a position above 75
Level (Low, Medium, High)	A label for position: Low is below 25, Medium is 25 to 75, High is 75 or above. It shows position only, not whether an appliance is efficient or wasteful
Random sample	Units chosen by chance, so the sample is expected to look like the whole group
Natural groups (clusters)	Separate kinds of homes, for example a group of homes that use a lot of electricity and a group of homes that use very little. None were found in this data
Simulated data	Data produced by a computer program, not measured in real homes
Flat rate	One price per kWh at every hour of the day
Usage charge	The part of the electricity price that depends on how much energy is used, in cents per kWh
Daily supply charge	A fixed daily fee for being connected, whatever the use. It is not included in this report

Term	Meaning
GST	Goods and services tax. The price used here includes GST
Default Market Offer (DMO)	A reference electricity price set each year by the Australian Energy Regulator (AER). It is used here as an example rate
Smart meter	An electronic meter that records energy use at short intervals
Illustration (not a forecast)	A what-if calculation that shows how a change would affect cost. It does not predict what will happen
Pilot	A small trial, here on 10 to 20 units, before extending to the whole portfolio

C. Method and assumptions

- Portfolio: 50 of 500 homes, random sample (the list of homes is saved), files portfolio50_by_appliance.csv, portfolio50_by_home.csv and cost_assumptions.csv.
- Rate: $0.33 per kWh, rounded from the AER's 33.14 c/kWh, including GST, usage charge only.
- Tools: Python (pandas, NumPy) for the analysis; a Power BI dashboard for the interactive view.

D. Sources

- Australian Energy Regulator, Default Market Offer, 2026-27 final determination, chapter 2 (prices and tariff caps, including GST).
- Australian Government, YourHome: Heating and cooling (thermostat settings and the cost of each extra degree).
- Smart Home Energy Consumption dataset, Kaggle (mexwell): https://www.kaggle.com/datasets/mexwell/smart-home-energy-consumption